

Trust-Tiered Knowledge Resolution and Adaptive Multi-Criteria Scoring for Localized Cold-Start Recommendations on Tasco Maps

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Local search and recommendation engines on digital map platforms are heavily constrained by spatial proximity, temporal availability, and database accuracy. In rapidly shifting urban ecosystems, keeping point-of-interest (POI) attributes fresh is a major operational challenge. According to the iPOS 2024 report, Vietnam has 323,010 food and beverage (F&B) outlets, but experiences a high churn rate with over 30,000 closures in a single half-year, making manual database curation (requiring 40 person-years per pass) unsustainable. This paper introduces a trust-tiered knowledge resolution and adaptive recommendation framework designed for the Vietnamese F&B landscape on the Tasco Maps platform. The full production architecture targets a five-tier trust registry that would ultimately incorporate merchant point-of-sale data (MISA CukCuk) and vehicle dwell telemetry from the VETC mobility network (2.8 million vehicles); this paper implements and evaluates the three tiers realizable without a live merchant or telemetry partnership — a licensed-source tier (Apify Google Places, confidence 0.80), an agentic web-scraping tier (TinyFish/Foody, confidence 0.60), and a social-buzz signal tier (TikTok/Facebook, engagement-weighted with exponential recency decay) — resolved by a highest-confidence consensus rule and validated on a real 537-POI corpus spanning Th Đức, Quận 2, Làng Đa Hòa and Khu Công Nghệ Cao, alongside a 30-POI synthetic benchmark used for anti-hallucination testing. For online serving, we implement a sequential two-stage pipeline: (1) a Hard-Constraint Gate that enforces Vietnamese dietary preferences, geographic bounds, and opening-hours honesty traps, and (2) a Multi-Criteria Soft-Scoring stage that renormalizes weights for missing signals. A repeat-vs-exploration mode-switching extension and live VETC/MISA data integration are specified as the production roadmap but are not yet backed by live data feeds.

CCS Concepts: • **Information systems** → **Recommender systems**.

Additional Key Words and Phrases: Cold-Start Recommendation, Tasco Maps, VETC Telemetry, iPOS Vietnam F&B, Trust Resolver, Detour Decay, Weight Renormalization

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1 Introduction

Digital map platforms are essential for local discovery, navigation, and location-based commerce. While global platforms like Google Maps excel at global scale and breadth, they frequently suffer from lack of depth, unverified metadata, and black-box ranking in localized markets. This limitation is particularly prominent in the Vietnamese food and beverage

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(F&B) industry. According to the iPOS Vietnam 2024 F&B Market Report [?], the country has approximately 323,010 active outlets. However, the industry suffers from a high churn rate: over 30,000 outlets closed in the first half of 2024 alone, indicating a semi-annual churn rate of approximately 10%.

To keep a map platform’s POI database accurate under such rapid decay, manual database curation is operationally unfeasible. At an estimated 15 minutes of human verification per POI, a single pass across 323,010 outlets requires 80,750 hours (equivalent to 40 person-years of manual labor). While large language models (LLMs) can scale extraction, naive LLM extraction frequently fabricates (hallucinates) missing attributes, causing brand damage.

This paper presents the design and formulation of a trust-tiered knowledge resolution and adaptive recommendation engine implemented for the next-generation digital map, **Tasco Maps**. The platform leverages two major local data alliances: (1) **MISA CukCuk** [?], a leading merchant POS provider in Vietnam with over 50,000 restaurants, and (2) **VETC** [?], a nationwide electronic toll collection network covering 2.8 million active vehicles. By combining merchant point-of-sale menus and real-time vehicle telemetry, the framework builds a self-updating, high-fidelity knowledge graph.

Our recommendation engine operates as a sequential pipeline: **Hard-Constraint Gating followed by Soft-Scoring**. The Hard-Constraint Gate prevents semantic-matching engines from silently relaxing constraints, addressing “honesty traps” (e.g., verifying if a requested dietary option actually exists). The Soft-Scoring stage ranks surviving candidates using a multi-criteria linear formulation incorporating semantic relevance, geographic decay, POI quality, and persona/localness priors. We further specify, but do not yet operate on live data, an extension in which the weights adapt dynamically based on travel frequency, separating repeat commuters from exploratory travelers, following the food delivery paradigm analyzed by Wang et al. [?].

1.1 Implementation Status

Because two of the five trust tiers depend on data partnerships (merchant POS, vehicle telemetry) that are still being negotiated, we distinguish throughout this paper between the *target production architecture* and what is *implemented and empirically validated* in the present system. Table ?? summarizes this split; components marked PROPOSED are fully specified (weights, data contracts, trigger conditions) but await a live data feed, while components marked IMPLEMENTED are running code, exercised against either the 30-POI synthetic honesty benchmark or the real 537-POI Th Đức corpus described in Section ??.

2 Related Work

We build on cold-start POI recommendations, spatial detour modeling, and situation-aware systems.

2.1 Cold-Start Recommendation and LLM Priors

Collaborative filtering fails when items or users have zero historical interactions. Ju et al. [?] introduced the Language-Model Prior (LM-Prior) framework, demonstrating that zero-shot LLM representations of item metadata can establish a strong baseline prior. For specialized discovery, KALM4Rec [?] leverages retrieval-augmented LLM architectures to extract and verify query keywords. In Vietnamese contexts, morphological segmenters like HuTieuBERT by Dinh et al. [?] are critical for parsing localized dish names (e.g., “bún ch”) and preventing entity-matching failures. Furthermore, processing noisy social media data (e.g., reviews, slang, and dialect expressions extracted from platforms like TikTok

Table 1. Implementation status of each system component

Component		Status
Hard-constraint (diet/city/price/hours)	gate	Implemented
Anti-hallucination (unknown-entity, diet)	traps empty-	Implemented
Soft-score (semantic/geo/quality/persona/rating-popularity/localness/anti-luxury)		Implemented
UGC contribution + menu OCR		Implemented
On-demand (/v1/enrich, Apify Google Places)	enrichment	Implemented
Trust resolver (Apify), Tier 3 (Tiny-Fish/Foody), Tier 4a (Tier 4b (TikTok/Facebook))	— Tier 3 4a 4b	Implemented, validated on real data
Social buzz (engagement-weighted, recency decay)	score	Implemented
Merchant POS tier (MISA CukCuk, Tier 1)		Proposed
Driver telemetry (VETC, Tier 2)	tier	Proposed
Parking as a hard constraint for car travelers		Proposed
Repeat-vs-explore (switching, crowd occupancy, route detour decay)	weight	Proposed (algorithm specified; exercised only via seeded demo trip context, no live telemetry)

or Facebook) requires robust domain-specific language models. We identify ViSoBERT, proposed by Nguyen et al. [?] as a pre-trained Transformer optimized for Vietnamese social media text, as the intended model for aspect-based sentiment extraction over the review corpus; at present, our implemented Tier 4b pipeline (Section ??) reduces social signals to an engagement-and-recency buzz score without yet performing sentiment or aspect extraction, and per-POI review sentiment remains an open pass-through field awaiting this integration.

2.2 Geographical POI Recommendation

POI recommendation is fundamentally shaped by geography. GeoMF by Lian et al. [?] models spatial regions within matrix factorization by applying spatial decay. In highway or

corridor environments, routing recommenders must account for detour paths rather than straight-line distance. Dynamic spatial routing structures (e.g., GETNext by Yang et al. [?]) model trajectory transitions, but their parameter size causes overfitting on small localized datasets, highlighting the need for parametric decay functions that scale with detour times.

2.3 Situation-Aware Repeat vs. Explore Recommendations

User preferences fluctuate between familiar routines and novel exploration. Wang et al. [?] studied food consumption patterns and proposed dual-model architectures (RepRec and ExpRec) to separate repeat orders from exploratory choices, showing that unified recommendation pipelines underperform when situation-aware contexts are ignored.

3 System Architecture and Preference Channels

The system separates offline background knowledge resolution from online real-time serving, ensuring low latency and resilience during network drops. The architecture features a five-tier registry that ingests and resolves data through a multi-trigger loop.

3.1 Multi-Trigger Lifecycle Loop

Rather than treating the database as static, the system runs a multi-trigger lifecycle loop governed by a set of triggers $\mathcal{T} = \{T_1, T_2, T_3, T_4, T_5, T_6\}$:

- **Bootstrap (T_1):** Initializes the database by ingesting baseline files and OpenStreetMap (OSM) nodes.
- **Gap-Detect (T_2):** Triggered when a POI's quality score $Q < 0.75$ or key attributes (e.g., operating hours) are absent. This triggers a localized agent run to enrich the profile.
- **TTL Expiry (T_3):** Triggered when the age of a resolved field exceeds its tier-specific Time-To-Live (τ), prompting a target refresh of the expired source: $\tau_1 = 180\text{d}$ (Tier 1), $\tau_2 = 90\text{d}$ (Tier 2), $\tau_3 = 30\text{d}$ (Tier 3), $\tau_{4a} = 14\text{d}$ (Tier 4a).
- **Business Event (T_4):** Triggered when a merchant uploads a menu image, initiating zero-shot OCR and structured menu extraction.
- **Driver Telemetry (T_5):** Triggered when GPS telemetry detects a vehicle passing and dwelling at a POI for ≥ 25 minutes, prompting a 1-tap validation.
- **User Query (T_6):** Triggers the online search and recommendation pathway.

3.2 Preference Input Channels

To construct personalization models with minimal user effort, the system aggregates user preferences through four complementary channels:

- (1) **Trip Context (CTX):** Fully passive channel extracting context from the active route navigation state, including the timestamp (t_{now}), arrival ETA (t_{arrival}), and vehicle class (e.g., cars require parking).
- (2) **Spatial-Temporal Autocomplete:** Captures query intents based on character prefixes combined with spatial-temporal priors. For example, the query prefix “ca” suggests coffee shops (“cà phê”) at 7:00 AM in urban zones, but shifts to fish soup (“cá”) at 12:00 PM on a highway corridor. This addresses the 17.9% intent variance observed by Li et al. [?] (Baidu MST-PAC).
- (3) **Onboarding Chips:** A one-time 10-second setup capturing critical hard constraints (allergens, dietary restrictions, or family travel) where incorrect predictions pose safety or high-dissatisfaction risks.

- (4) **Implicit Learned Behavior:** Collects telemetry-based signals such as vehicle dwell times (indicating repeat customers), navigation requests, and post-trip 1-tap feedback.

4 Trust-Tiered Knowledge Resolution

Data is ingested from multiple sources, each mapped to a specific trust tier $t \in \{1, 2, 3, 4a, 4b\}$, as shown in Table ?? . Each tier carries a static confidence weight $C_t \in [0, 1]$ and a time-to-live τ_t (in days).

Table 2. Trust Tier Hierarchy

Tier	Source Type	Confidence (C_t)	TTL (τ_t)	St
Tier 1	Merchant Verified (MISA CukCuk POS / OCR)	0.95	180 days	Pr
Tier 2	Driver Telemetry (VETC vehicle dwells)	0.85	90 days	Pr
Tier 3	Licensed/Structured API (Apify Google Places)	0.80	30 days	Im
Tier 4a	Agentic Web Scraping (TinyFish/Foody)	0.60	14 days	Im
Tier 4b	Social Buzz Signals (TikTok/Facebook)	N/A	14-day half-life	Im

Tiers 3, 4a, and 4b are the ones we could realize without a live merchant or telemetry partnership: Tier 3 uses Apify’s Google Places actor (real GPS coordinates, ratings, review text, opening hours); Tier 4a uses the TinyFish web agent to extract structured listings from schema-varying pages such as Foody; Tier 4b aggregates TikTok video and Facebook page mentions into the social buzz score of Section ?? . Tiers 1 and 2 require, respectively, a merchant POS data-sharing agreement and a live VETC telemetry feed, neither of which is available at the time of writing; they are included in the registry design so the resolver’s tier-priority and TTL rules extend to them without modification once those partnerships exist.

Let $\mathcal{V}_f = \{(v, t, \Delta t)_j\}$ be the set of candidate values extracted for a field f from active sources, where v is the value, t is the source’s trust tier, and Δt is the data age. The Resolver determines the published value v_f^* and its confidence $\text{conf}(f)$ through the following consensus rules:

- (1) **TTL Filtering:** Remove all candidates where the elapsed time $\Delta t > \tau_t$.
- (2) **Highest Tier Selection:** Select candidates corresponding to the highest surviving tier $t_{\max}$. If multiple distinct values exist within the same tier, the most recently updated value (lowest Δt) is selected.
- (3) **Consensus Constraint:** For Tiers 2 (driver confirmed) and 4a (agentic facts), a candidate value is only valid if supported by a consensus of $N \geq 3$ independent sources. A single agentic source cannot override fresh licensed data.
- (4) **Absence Gating:** If no candidates survive, the field is marked as *absent*. No default values are fabricated. This prevents recommendation hallucinations under strict search queries (such as queries for non-existent entities like “Crystal BBQ”).
- (5) **Full History Retention:** Retains the entire candidate list to provide source provenance for the RAG assistant (/assistant).

The resolver preserves the winning tier’s confidence as $\text{conf}(f) = C_{t_{\max}}$, which is propagated to the ranking stage. Tier 4b sources bypass this resolver and feed directly into the buzz score of Section ?? rather than competing for a field value.

Of these five rules, our implemented resolver (validated on the 537-POI corpus of Section ??) realizes Highest-Tier Selection and Absence Gating as written. TTL Filtering and Full History

Retention are specified but not yet enforced by the current single-pass resolver, which keeps only the winning value per field rather than the full candidate history. The Consensus Constraint is likewise not yet enforced quantitatively: with only Tier 3 and Tier 4a populated for most POIs (rarely more than two independent sources per field today), we cannot yet require $N \geq 3$ agreement without simply discarding most fields. Until Tier 1/2 partnerships and broader Tier 4a coverage supply enough independent sources, single-source Tier 4a values are accepted at their tier’s confidence rather than gated on consensus — a scoping decision we flag explicitly rather than silently understate.

5 Mathematical Formulation and Online Serving

Let $U = (q, l, s, d, p_c, t, \mathcal{H})$ represent the user context, where q is the query string, $l = (\phi_u, \lambda_u)$ is the location, $s \in \mathcal{S}$ is the user segment, $d \in \mathcal{D}$ denotes dietary requirements, p_c is the maximum price threshold, t is the timestamp, and $\mathcal{H}$ is the user’s travel history on the current route. Let R denote a candidate restaurant POI.

5.1 Stage 1: Hard-Constraint Gating

To prevent semantic matching from violating physical constraints, candidate POIs are filtered using a binary indicator $H(R, U) \in \{0, 1\}$:

$$H(R, U) = \prod_{k \in \mathcal{K}_{\text{active}}} h_k(R, U) \quad (1)$$

where $\mathcal{K}_{\text{active}}$ represents the active constraint set, evaluating:

- **Dietary Match (h_{diet}):** Checks if dietary tags match the resolved properties of R or its menu items:

$$h_{\text{diet}}(R, U) = \begin{cases} 1 & \text{if } d = \emptyset \text{ or } \exists m \in \mathcal{M} \text{ s.t. } \text{status}(m, d) \geq 0.60 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

- **Geographic Max Boundary (h_{geo}):** Validates if the physical distance is within the query radius limit.
- **Operating Hours (h_{time}):** Evaluates opening hours at the estimated arrival time $t_{\text{arrival}} = t_{\text{now}} + \text{ETA}$, rather than the query time t_{now} .
- **Price Ceiling (h_{price}):** Rejects R if the average price $P_a > p_c$ or if all matched dishes exceed p_c .
- **Parking Requirement (h_{parking}), proposed:** Designed to escalate to a hard constraint when vehicle class is ‘car’. The underlying signal exists — our real-world corpus (Section ??) captures a car-specific parking field per POI, distinct from generic motorbike parking — but wiring it into the live query-time gate is not yet deployed.

If the gated set $\mathcal{R}_{\text{gate}} = \{R \mid H(R, U) = 1\} = \emptyset$, the system returns an explicit empty state rather than relaxing constraints silently, addressing honesty traps (such as requests for Halal food in districts with zero certified options).

5.2 Stage 2 (Implemented): Multi-Criteria Soft-Scoring

For each candidate $R \in \mathcal{R}_{\text{gate}}$, the deployed engine computes:

$$S(R, U) = w_1 S_{\text{sem}} + w_2 S_{\text{geo}} + w_3 S_{\text{qual}} + w_4 S_{\text{pers}} + w_5 S_{\text{rp}} + w_6 S_{\text{loc}} - w_7 S_{\text{lux}} \quad (3)$$

with $(w_1, \dots, w_6) = (0.30, 0.20, 0.15, 0.15, 0.10, 0.10)$ — summing to 1 over the six positive components — and a subtracted anti-luxury penalty $w_7 = 0.25$. The components are:

- (1) **Semantic Relevance** (S_{sem}): lexical token overlap between the query and the POI's name/menu/review index, boosted when a named dish matches.
- (2) **Geographic Decay** (S_{geo}): $\exp(-d/2\text{km})$ over the haversine distance d to the user.
- (3) **Information Quality** (S_{qual}): the precomputed POI completeness score.
- (4) **Preference Match** (S_{pers}): average match over the segment/diet/price dimensions supplied in U .
- (5) **Rating-Popularity** (S_{rp}): $(\text{rating}/5 + \text{popularity}/100)/2$.
- (6) **Localness** (S_{loc}): a boost for affordable, high-quality local (“bình dân”) eateries, per the Tasco brief's preference for small authentic venues over international chains.
- (7) **Luxury Penalty** (S_{lux}): a subtracted penalty applied to premium-tier venues.

Each result exposes this breakdown as a `why{}` object plus a Vietnamese **reasoning** string (e.g. “cách bn 0.7km · giá bình dân”), making the ranking auditable rather than a black box.

5.3 Stage 2 (Proposed): Target Production Extension

The production architecture extends Section ?? with three signals that require data sources not yet live — route detour time, vehicle dwell history, and real-time crowd occupancy — plus a resolver-confidence-weighted preference match and a repeat-vs-explore mode switch. We specify the full target formulation so it can be adopted once the corresponding tier (Table ??, Tiers 1–2) is connected:

$$\tilde{S}(R, U) = \sum_{i \in \mathcal{I}} w_i S_i(R, U), \quad \sum_{i \in \mathcal{I}} w_i = 1, \quad S_i(R, U) \in [0, 1] \quad (4)$$

- (1) **Preference Match, confidence-weighted** ($w = 0.20$): $S_{\text{match}}(R, U) = \text{MatchRatio}(q, \mathcal{A}) \cdot \text{conf}(\mathcal{A})$ — the persona match of Section ?? scaled by the resolver's confidence in the matched attributes (Section ??). Because Tiers 3/4a/4b are live, $\text{conf}(\mathcal{A})$ is computable today; only the wiring into the deployed scorer is pending.
- (2) **Detour Decay** (S_{route} , $w = 0.15$): $\exp(-\Delta t_{\text{detour}}/5\text{min})$ — requires live routing (Valhalla), not yet integrated.
- (3) **Behavioral History** (S_{behavior} , $w = 0.15$): visit frequency on the current corridor, drawn from vehicle telemetry $\mathcal{H}$ — requires Tier 2 (VETC); currently exercisable only against seeded demo trip contexts, never a live feed.
- (4) **Crowd Occupancy** (S_{crowd} , $w = 0.05$): $1 - \text{occupancy}(t_{\text{arrival}})/100$ — requires a live popular-times feed.

Social Buzz (S_{buzz}) is deliberately excluded from this proposed list: unlike the other extensions, it *is* implemented and evaluated on the real corpus, described next.

5.4 Social Buzz Score (Implemented)

Tier 4b signals (TikTok videos, Facebook pages) are matched to a POI by normalized name overlap and reduced to a single buzz score rather than treated as individual facts, consistent with the resolver rule that Tier 4b never overrides a structured field (Section ??). For a POI R matched by signal set Σ_R , each signal σ 's engagement is indexed as

$$\text{idx}(\sigma) = \sum_k \omega_k e_k(\sigma) \quad (5)$$

where $e_k(\sigma)$ is the raw engagement count on channel k (likes/comments/shares/views for TikTok; likes/followers/rating-count for Facebook) and ω_k up-weights channels indicating deliberate engagement over passive exposure (comments and shares weighted 2–3× likes;

views weighted $0.05\times$). Each signal’s contribution decays exponentially with a 14-day half-life,

$$\text{decay}(\sigma) = 0.5^{\text{age}(\sigma)/14}, \quad (6)$$

and the buzz score sums decayed engagement across matched signals, squashed into $[0, 1]$ so a single viral post cannot dominate:

$$S_{\text{buzz}}(R) = 1 - \exp\left(-\frac{1}{\kappa} \sum_{\sigma \in \Sigma_R} \text{idx}(\sigma) \cdot \text{decay}(\sigma)\right), \quad (7)$$

with scale constant κ calibrated against observed engagement magnitudes in our corpus. Following the resolver’s absence-gating principle (Section ??), a POI with no matched social mention receives $S_{\text{buzz}} = (\text{absent})$, never a fabricated zero — of the 537 POIs in our corpus (Section ??), only a minority are matched, since name-matching against short-form video captions and page titles is deliberately conservative to avoid false attribution.

5.5 Adaptive Renormalization for Missing Signals

Online requests often lack GPS context (denied permissions) or behavioral logs. To prevent score distortion, we exclude missing features from the active set $\mathcal{I}_{\text{active}} \subseteq \mathcal{I}$ and renormalize the weights:

$$S(R, U) = \sum_{i \in \mathcal{I}_{\text{active}}} w_i^* S_i(R, U) \quad \text{where} \quad w_i^* = \frac{w_i}{\sum_{j \in \mathcal{I}_{\text{active}}} w_j} \quad (8)$$

This rule is implemented for Section ??’s deployed formula (e.g. dropping S_{geo} when the user denies location) and specified identically for the proposed extension of Section 5.2 once its signals go live.

5.6 Context-Aware Mode Switching (Proposed)

Following the repeat-explore behavioral dichotomy in local food delivery studied by Wang et al. [?], the production design calls for dynamic mode-switching over the Section 5.2 weight vector, based on route familiarity (a route is *familiar* if user history contains ≥ 3 trips within the past 90 days):

- **Repeat Mode (Familiar Routes):** Prioritizes user habit and crowd status, decreasing the semantic weight and ignoring temporary social buzz:

$$w = [0.25, 0.20, 0.15, 0.25, 0.10, 0.05, 0.00]$$

representing $[w_{\text{sem}}, w_{\text{mat}}, w_{\text{rout}}, w_{\text{beh}}, w_{\text{qual}}, w_{\text{cro}}, w_{\text{buz}}]$.

- **Explore Mode (New Routes):** Prioritizes semantic discovery and trending social buzz:

$$w = [0.35, 0.20, 0.15, 0.05, 0.10, 0.05, 0.10]$$

As with the rest of this extension, mode-switching is specified and ready to enable but is not yet driven by a live telemetry feed.

5.7 Vietnam-Specific Adjustments: Localness and Anti-Luxury Bias (Implemented)

Under the Tasco Maps platform guidelines, there is a preference for small, local, and authentic eateries (“bình dân”) over high-end international luxury chains. This is already realized in Section ??’s deployed formula as the S_{loc} and S_{lux} terms: $S_{\text{loc}}(R) \rightarrow 1$ when the price tier is budget and POI quality is high, and $S_{\text{lux}}(R) \rightarrow 1$ when the price level is premium, so that $w_6 S_{\text{loc}} - w_7 S_{\text{lux}}$ shifts the ranking toward affordable, well-regarded local venues without a separate post-processing pass.

5.8 LLM Rerank and Explainer (Proposed)

Following the classical-plus-neural recommendation architectures described by DoorDash [?], the production design passes the top-20 candidates from Stage 2 to an LLM reranking step: the classical scorer handles scaling and candidate pruning, while the LLM incorporates contextual semantics and generates a natural-language explanation (e.g., “4 minutes detour from route, has car parking [owner verified], positive reviews on TikTok”) in Vietnamese. This stage was the first item cut under the hackathon’s time budget and remains unimplemented; Section ??’s `why{}/reasoning` output already gives an auditable, template-generated Vietnamese explanation without it, which is what is currently deployed.

5.9 Real-World Validation Corpus (Implemented)

The 30-POI benchmark used for honesty-trap evaluation is synthetic by design (Section ??), so it cannot demonstrate the trust resolver operating on genuinely heterogeneous, real-world data. To validate that separately, we built a second corpus of 537 real restaurants across Th Đức, Qun 2, Làng Di Hc, and Khu Công Ngh Cao — areas outside the synthetic benchmark’s coverage — using exactly the Tier 3/4a/4b pipeline of Table ??: Tier 3 via Apify’s Google Places actor (real GPS coordinates, ratings, opening hours, and up to 15 reviews per place); Tier 4a via the TinyFish web agent extracting structured listings from Foody.vn; and Tier 4b via TikTok video search and Facebook business-page search, matched to a resolved POI by normalized name overlap. Every field carries the $\{value, source, confidence, fetched_at\}$ provenance tuple of Section ??, and the resolver’s highest-confidence-wins rule is exercised for real on any POI independently surfaced by both Tier 3 and Tier 4a. The compiled corpus additionally carries a car-specific parking flag per POI — distinct from the generic motorbike parking Vietnamese listings usually report — motivating the proposed h_{parking} hard constraint of Section ??, and the S_{buzz} score of Section ??.

6 Discussion and Future Work

The trust-tiered resolver provides a mechanism to manage the trade-off between coverage and accuracy, validated today across the three tiers we could populate without a data partnership (Tier 3 Apify, Tier 4a TinyFish/Foody, Tier 4b social buzz). Closing the gap to the full five-tier design is primarily a data-access problem, not an algorithmic one: Tier 1 requires a MISA CukCuk merchant data-sharing agreement, and Tier 2 requires a live VETC telemetry feed, neither of which exists yet. Once either is available, the resolver’s tier-priority and TTL rules extend to it unmodified, and the quantitative $N \geq 3$ consensus constraint becomes enforceable once enough independent sources populate a given field.

Similarly, the repeat-vs-explore mode switch and the route/behavior/crowd terms of the proposed Stage-2 extension (Section 5.2) are specified with concrete weights but currently only exercisable against seeded demo trip context, not live telemetry — so we report them as a designed, ready-to-enable extension rather than an evaluated one. A logical next step, once real usage logs exist, is to replace the statically defined Repeat/Explore weight vectors with a meta-learning layer that interpolates weights from continuous context features. Additionally, while the linear model is well suited to cold-start environments for its interpretability and audibility (the `why{}` breakdown), scaling to millions of POIs will eventually motivate a deep architecture; the Hard-Constraint Gate is designed to continue serving as a pre-filtering layer ahead of any such model, preserving the honesty guarantees validated in Section ??.

7 Conclusion

We presented a two-stage localized recommendation framework combining trust-tiered knowledge resolution with multi-criteria soft-scoring, and we have been explicit about which parts of it are running code versus specified roadmap. The implemented system — hard-constraint gating with validated anti-hallucination traps, a seven-factor soft score with auditable `why{}` explanations, a three-tier resolver spanning licensed, agentic, and social-buzz sources, and an engagement-weighted buzz score with honest absence-gating — is validated on both a 30-POI synthetic honesty benchmark and a real 537-POI corpus assembled via web-agent scraping. The proposed production extension — merchant POS and vehicle-telemetry tiers, detour-aware routing, crowd occupancy, and repeat-vs-explore mode switching — is fully specified so it can be adopted without redesign once the corresponding data partnerships and live feeds exist. We believe this implemented/proposed split, rather than a single undifferentiated architecture diagram, is itself a useful contribution for teams building localized recommenders against data sources that arrive incrementally.

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